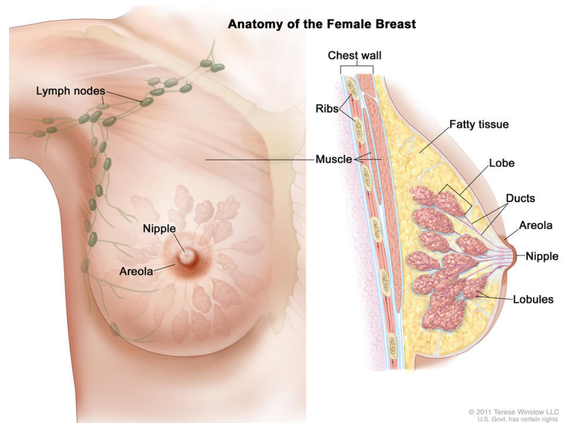
NATIONAL CANCER INSTITUTE

What Is Breast Cancer?

Breast cancer is cancer that starts in the breast. It can start in one or both breasts. Breast cancer happens when cells in the breast grow without control, creating a mass called a tumor that may spread elsewhere in the body. Breast cancer mostly affects females aged 45 and older, but anyone with breasts can get breast cancer. It is rare in children and males.



The female breast contains lobes, lobules, and ducts that produce and transport milk to the nipple. Fatty tissue gives the breast its shape, while muscles and the chest wall provide support. The lymphatic system, including lymph nodes, filter lymph and store white blood cells that help fight infection and disease.

Credit: © Terese Winslow

Enlarge Image

Breast cancer can form in:

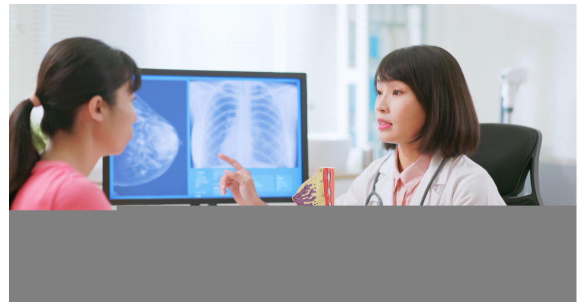
- **Glandular tissue.** This tissue includes milk glands, milk ducts, and lobules. Lobules are tiny glands that make milk. After a baby is born, breast milk flows from the lobules through thin tubes (ducts) to the nipple. Cancers that start in the ducts are called ductal cancers. Cancers that start in the lobules are called lobular cancers. Most breast cancers are ductal cancers.
- **Fibrous tissue and fatty tissue (stroma).** These tissues fill the spaces between the lobules and ducts and hold breast tissue in place. They give breasts their shape and size. A type of breast tumor called phyllodes tumor can start in the stroma.
- **The nipple.** This is the small, raised area in the center of the breast that contains duct openings through which milk can leave the breast. A type of breast cancer called Paget disease of the breast can start in the nipple.
- **Blood vessels and lymph vessels.** Inflammatory breast cancer occurs when cancer cells block lymph vessels in the skin of the breast. Angiosarcoma can start in the cells that line blood vessels and lymph vessels in the breast.

There are many types of breast cancer, depending on where it begins and the extent of spread. When the abnormal cells are within the lobules or ducts and have not spread to other tissues in the breast, it is called carcinoma in situ. Invasive cancers have spread into surrounding breast tissue and can spread to nearby lymph nodes or other organs throughout the body. Most breast cancers are invasive.

NIH NATIONAL CANCER INSTITUTE

Types of Breast Cancer

There are many types of breast cancer. The types differ based on several factors, such as which cells in the breast become cancer, whether the cancer has spread from where it first formed, and whether the cancer has certain features that affect treatment options.



Talk to your doctor to find out what type of breast cancer you have and how it is used to plan the best treatment for you.

Credit: iStock

Invasive or infiltrating ductal carcinoma begins in the cells that line the milk ducts and has spread beyond where it first formed. It is the most common breast cancer diagnosis. Learn more about Breast Cancer.

Invasive lobular carcinoma begins in the cells that line the breast glands that make milk, called lobules, and has spread beyond where it first formed. It grows more slowly and is less common than invasive ductal carcinoma. Lobular carcinoma is more often found in both breasts than are other types of breast cancer.

Inflammatory breast cancer is a rare form of breast cancer in which cancer cells block lymph vessels in the skin of the breast. This type of breast cancer has a high risk of recurrence. It is called inflammatory because the affected breast often looks swollen and red, or inflamed.

Triple-negative breast cancer is a form of breast cancer in which the cancer cells lack features that are common in breast cancer, including hormone receptors and a protein called human epidermal growth factor receptor 2 (HER2). This form of breast cancer has a higher risk of recurrence than most other forms of breast cancer.

Metastatic breast cancer, also called stage 4 breast cancer, is breast cancer that has spread from the breast to another part of the body. Metastatic spread occurs when breast cancer cells break away from the original tumor and travel through the lymph system or blood to other sites in the body.

Ductal carcinoma in situ (DCIS) forms in the cells that line the milk ducts but has not spread beyond where it first formed. It may also be called noninvasive breast cancer, intraductal carcinoma, or stage 0 breast cancer.

Paget disease of the breast is a rare type of cancer that involves the skin of the nipple and the areola.

Phyllodes tumor (also called cystosarcoma phyllodes of the breast, or CSP) is a rare type of breast tumor that starts in the connective tissue of the breast. Most phyllodes tumors are not cancer. Learn more at Benign and Precancerous Breast Lumps and Conditions.

Lobular carcinoma in situ (LCIS) forms in the cells that line the breast glands that make milk, called lobules, but has not spread beyond where it first formed. LCIS is not breast cancer but increases the risk of developing breast cancer in the future. Learn more at Benign and Precancerous Breast Lumps and Conditions.

Molecular subtypes of breast cancer

Molecular subtypes of breast cancer are defined by whether they have hormone receptors, HER2 protein, or other biomarkers. Examples of molecular subtypes of breast cancer include triple-negative, luminal A, luminal B, and HER2-positive. Learn about how these subtypes are diagnosed and how they affect treatment at Tests for Breast Cancer Biomarkers and Breast Cancer Treatment by Stage.

NIH NATIONAL CANCER INSTITUTE

Breast Cancer Causes and Risk Factors

Breast cancer is caused by certain changes in how breast cells function, especially how they grow and divide into new cells. A risk factor is anything that increases the chance of getting a disease. Some risk factors for breast cancer, like drinking alcohol, can be changed. However, risk factors also include things people cannot change, like your genetics, getting older, and your family history.

A family history of breast cancer in a first-degree relative (mother, daughter, or sister) increases your risk of breast cancer.

Credit: iStock

On This Page

Factors that can increase the risk of breast cancer

There are many risk factors for breast cancer, but most do not directly cause cancer. Instead, they increase the chance of DNA damage in cells that may lead to breast cancer. Learn more about how cancer develops at What Is Cancer?

Having one or more of these risk factors does not mean that you will get breast cancer and some people who get breast cancer do not have known risk factors. The most important risk factor for breast cancer, besides female biological sex, is increasing age.

Personal health history and breast conditions

The following personal factors and breast conditions have been linked to a higher risk of breast cancer:

- Having had breast cancer in the past.
- Having had ductal carcinoma in situ (DCIS) in the past.

- Having been exposed to high levels of radiation in the past, particularly at a young age, such as during treatment for childhood cancer.
- Having a history of certain kinds of breast changes. Learn more about precancerous breast conditions.
- Having dense breasts, a common condition that not only increases the risk of breast cancer but also makes it more difficult to detect breast cancer on mammograms. Learn more about dense breasts.
- Having excess body weight, especially after menopause.
- Taking combination (estrogen plus progestin) hormone replacement therapy for symptoms of menopause. Learn more about Menopausal Hormone Therapy and Cancer.
- Having been exposed to DES (a synthetic estrogen) before birth (in utero) or having taken it during pregnancy. Learn more about cancer risks linked to DES exposure.

Reproductive history

If your reproductive history leads to a longer exposure to natural estrogen, it can increase your risk of developing breast cancer. Reproductive factors that increase the length of time a woman's breast tissue is exposed to estrogen include:

- having a longer menstrual history (i.e., having early menarche, later menopause)
- being at an older age at first birth
- never having carried a pregnancy to term
- never having breastfed (among women who have given birth)

Genetics and family history

The following genetics and family history factors have been linked to a higher risk of breast cancer:

- having a harmful change (mutation or pathogenic variant) in a gene associated with breast cancer, such as:
  - *BRCA1* and *BRCA2* (learn more at BRCA Gene Changes: Cancer Risk and Genetic Testing)
  - *PALB2*
  - *CHEK2*
  - *TP53* (Li-Fraumeni syndrome)
  - *CDH1*

    - *PTEN* (Cowden syndrome and PTEN hamartoma tumor syndrome)
    - *STK11* (Peutz-Jeghers syndrome)
    - *ATM*
- having a family history of breast cancer

Genetic counselors and other specially trained health professionals can help people with breast cancer or with a family history of breast cancer understand:

- the likelihood that they have a genetic risk factor for breast cancer based on their personal history and their family medical history
- their options for genetic testing for changes in the *BRCA1*, *BRCA2*, and other genes that increase the risk of breast cancer
- the risks and benefits of learning genetic information
- how to cope with their genetic testing results
- how to discuss the results with family members

Learn more about genetic counseling and genetic testing to assess your risk of breast cancer and other cancers at Genetic Testing for Inherited Cancer Risk.

Behaviors

The following behaviors have been linked to a higher risk of breast cancer:

- Drinking alcohol. The more alcohol you drink the greater your risk of breast cancer. Learn more about Alcohol and Cancer Risk.
- Not being physically active, especially after menopause. Learn more about Physical Activity and Cancer.

Race

The incidence of breast cancer varies with race. In the United States, White women have the highest incidence of breast cancer, whereas Black women have the highest death rate.

Potential risk factors that require more study

Some studies have shown that women who are current or recent users of hormonal birth control pills (also called oral hormonal contraceptives) may have a slight increase in breast cancer risk. Other studies have not shown an increased risk of breast cancer in women using birth control pills. Learn more about Oral Contraceptives and Cancer Risk.

Scientists are also studying whether exposure to certain chemicals in the environment may increase a person's risk of breast cancer. Studies of this kind can be difficult to conduct and interpret for many reasons, making it hard to know which chemicals, if any, may increase the risk of breast cancer.

Understanding your risk of breast cancer

Everyone with breast tissue has some risk of breast cancer, but certain risk factors may increase your risk.

On average, women have a 1 in 8 (or 13 in 100) chance of developing breast cancer during their lifetime. If you have certain risk factors—for instance, you got your menstrual period at an early age, you were older when your first child was born, or you have a first-degree relative with breast cancer—you may be at higher risk. To help you understand your own risk, your doctor can use a risk calculator, such as NCI's Breast Cancer Risk Assessment Tool, which estimates people's risk of breast cancer based on these and several other clinical, reproductive, and medical history factors.

There are some risk factors that, on their own, confer a very high risk of breast cancer—60 in 100 or more. These include certain harmful genetic changes in genes such as *BRCA1*, *BRCA2*, *PALB2*, and *PTEN*; a strong family history of breast cancer (with multiple relatives diagnosed); and having had radiation to the chest before age 30.

Women who may be at increased risk of breast cancer can discuss possible screening or preventive measures with their doctors.

It is important to keep in mind that, even with the best available scientific information, doctors cannot precisely estimate an individual's risk. All estimates are subject to uncertainty. For example, someone might have a personal factor or exposure that influences their risk but that isn't included in the risk calculator. Also, scientific evidence can change over time, making earlier calculations less accurate. Estimates of risk should be considered best guesses based on the knowledge we have now.

Breast Cancer Prevention

Learn about behaviors and strategies that can reduce your breast cancer risk.

NIH NATIONAL CANCER INSTITUTE

Breast Cancer Signs and Symptoms

Signs and symptoms of breast cancer may include a lump or change in your breast. Signs and symptoms may vary, based on the type of breast cancer as well as how advanced the cancer is. However, early breast cancer often has no symptoms, which is why breast cancer screening is important.

A lump or change in the breast can be a symptom of breast cancer.

Credit: iStock

On This Page

What are the symptoms of breast cancer?

Although most breast changes are not cancer, it is important to check with your doctor if you notice unusual changes, including:

Breast lump, such as:

- a lump in or near your breast
- a lump under your arm
- thick or firm area (or mass) in or near your breast or under your arm
- a change in the size or shape of your breast

Nipple changes or discharge, such as:

- fluid or discharge that is not breast milk
- changes in the shape of the nipple or the direction it points or flattening of the nipple

Skin changes on your breast(s), such as:

- scaly or swollen skin on the breast, nipple, or areola
- redness or darkening of the skin on the breast, nipple, or areola
- itching or tingling in the nipple or areola
- general swelling on your breast; there may not be a lump
- skin rash
- dimples or puckering

Depending on your symptoms, your doctor may suggest that you have a diagnostic mammogram or another test to find out if the symptoms are due to cancer or to a benign (not cancer) condition. Learn more about How Breast Cancer Is Diagnosed.

Most breast changes are not a sign of breast cancer. Learn about Benign and Precancerous Breast Lumps and Conditions.

Breast Cancer Screening

Since early breast cancer does not often cause symptoms, screening mammograms are an important way to find breast cancer early. Learn about Breast Cancer Screening.

Is pain a symptom of breast cancer?

Breast cancer does not usually cause pain. Several conditions that are not cancer, such as breast cysts and hormonal changes before your period, can cause breast soreness or pain, as can certain medications. If you have breast pain that persists, it's important to see a doctor.

What are symptoms of metastatic breast cancer?

Breast cancer can sometimes spread beyond the breast, at which point it may be described as advanced breast cancer. It can spread to organs near the breasts, or it can metastasize (spread) through the blood or lymph system to distant sites, including the brain, bones, liver, lungs, or other organs.

If breast cancer is advanced or metastatic, it may cause signs and symptoms such as back pain, bone pain or bone fractures, shortness of breath, a constant dry cough, abdominal pain, swelling, jaundice, severe headaches, seizures, or vision changes. Symptoms depend on where the cancer has spread.

Learn more about Metastatic Breast Cancer, also called stage IV breast cancer.

What should I do if I find a breast change?

If you or your doctor find a breast lump or other breast change, keep in mind that breast changes are common and most are not symptoms of breast cancer. But it’s important to follow up with your doctor when you notice a breast change, even if you had a recent normal mammogram.

NATIONAL CANCER INSTITUTE

Breast Cancer Screening

Breast cancer screening is looking for breast cancer in people who do not have symptoms of the disease. This can help find breast cancer at an earlier stage. When breast cancer is found early, it may be easier to treat.

Breast cancer screening has been found to reduce deaths from breast cancer and is an important part of routine health care for women.

It is important to remember that your doctor does not necessarily think you have cancer if he or she suggests a screening test. Screening tests are done when you have no cancer symptoms.

Credit: iStock

On This Page

Tests used to screen for breast cancer

Several tests may be used to screen for breast cancer, with mammography being the standard test for most women. Some of these tests may also be used as part of diagnosing breast cancer.

Mammography

A mammogram is an x-ray picture of the breast. Learn more about mammography, including who should be screened and when, what to expect during a mammogram, and how results are reported at Mammograms.

Breast magnetic resonance imaging (MRI)

Breast MRI uses radio waves, a strong magnet, and a computer to create detailed pictures of the inside of the breasts. It does not involve radiation. If you are at high risk for developing breast cancer you may have breast MRI along with mammography. That is because MRI is more sensitive at detecting cancer than mammography and mammography alone may miss some cancers in high-risk women.

If you have dense breasts you may be offered breast MRI along with mammography. However, it is not known whether additional, or supplemental, screening with MRI leads to better health outcomes in women with dense breasts. Learn more about dense breasts.

Breast ultrasound

Breast ultrasound uses high-energy sound waves to look at tissues and organs inside the body. If you have dense breasts, you may be offered screening with ultrasound in addition to mammography. It is not known whether additional screening with breast ultrasound leads to better health outcomes.

Clinical breast exam

A clinical breast exam is a physical exam of the breast done by a health care provider. He or she will carefully feel the breasts and under the arms for lumps or anything else that seems unusual. A clinical breast exam alone is not an adequate screening test for breast cancer.

Breast self-exam

It is important for all women to be aware of how their breasts normally feel so that they can check with their doctors about any unusual changes. Even if you have recently had a normal mammogram, you should let your doctor know if you feel something unusual. However, breast self-exams are not an adequate breast cancer screening test on their own.

What are the benefits and harms of breast cancer screening?

All cancer screening can have both benefits and harms.

The benefit of breast cancer screening is that finding breast cancer by screening before it causes symptoms means treatment can start earlier and potentially be more effective.

The potential harms of breast cancer screening include:

- **False-positive results.** False-positive results can cause mental and emotional distress and require additional follow-up procedures, such as biopsies, that can themselves cause harm. False-positive results are more common among younger women, women with dense breasts, women who have had previous breast biopsies, women with a family history of breast cancer, and women who are taking estrogen (for example, menopausal hormone therapy).

- **Overdiagnosis and overtreatment.** Some cancers found during screening may not have caused problems during a woman's lifetime and would not have needed treatment. Finding these cancers earlier may not improve someone's health or help them live longer but will expose them to the harms of treatment.
- **Delayed diagnosis.** Diagnosis can be delayed due to false-negative results. False-negative results are more common among women with dense breasts. Diagnosis can also be delayed if someone with a recent normal screening test result does not check with their doctor about symptoms that could be from an interval breast cancer.
- **Radiation exposure.** Radiation from mammography can potentially cause cancer, although the risk is very low.

Learn more about the tradeoffs between benefits and harms of cancer screening in the Cancer Screening Overview.

NIH NATIONAL CANCER INSTITUTE

How Breast Cancer Is Diagnosed

If you have symptoms or screening test results that suggest breast cancer, your doctor will need to find out if they are due to cancer or some other cause. Your doctor may:

- ask about your personal and family medical history to learn more about your symptoms and possible risk factors for breast cancer
- do a physical exam, including a clinical breast exam, to feel for lumps or anything else that seems unusual

Depending on your symptoms and medical history, your doctor may recommend tests to find out if you have breast cancer. The following tests and procedures are used to diagnose breast cancer. If results of these tests show that you have breast cancer, they will also help you and your doctor plan treatment.

On This Page

Imaging tests to diagnose breast cancer

- A **diagnostic mammogram** is an x-ray picture of the breast that checks for breast cancer. It is used after a lump or other symptom of the disease has been found. It is also used to follow up on breast changes found during a screening mammogram. A diagnostic mammogram often involves taking more detailed x-ray pictures of the breast from different angles to check the abnormal area more closely.
- **Breast ultrasound** is a procedure that uses high-energy sound waves to make pictures of the inside of the breast.

- **Breast MRI** uses a powerful magnet, radio waves, and a computer to take detailed pictures of areas inside the breast.

Biopsy for breast cancer

You may have a biopsy if imaging tests find a lump or other abnormal area in the breast. In a biopsy, a surgeon removes cells or tissue so a pathologist can study them under a microscope. A biopsy is the only sure way to diagnose breast cancer.

An MRI of the breast is a procedure that uses radio waves, a strong magnet, and a computer to create detailed pictures of the inside of the breast. A contrast dye may be injected into a vein (not shown) to make the breast tissues easier to see on the MRI pictures. An MRI may be used with other breast imaging tests to detect breast cancer or other abnormal changes in the breast. It may also be used to screen for breast cancer in some people who have a high risk of the disease. *Note: The inset shows an MRI image of the insides of both breasts. Credit for inset: The Cancer Imaging Archive (TCIA).*

Credit: © Terese Winslow

Enlarge Image

The following types of biopsies may be used to check for breast cancer:

- **Fine-needle aspiration biopsy** uses a thin needle to remove tissue or fluid.
- **Core-needle biopsy** uses a wider needle to remove tissue samples, sometimes called cores. A small metal clip may be placed in the biopsy area to mark the spot for any future procedures.
- **Image-guided biopsy** is sometimes used for fine-needle aspiration biopsies and often used for core-needle biopsies. In this type of biopsy, ultrasound, mammography, or MRI are used to guide the needle to the area where the tissue needs to be checked. It is often used when the abnormal area is deep inside the breast or when the doctor cannot feel a lump or mass. When mammography is used to guide the needle, the procedure may be called a stereotactic biopsy.
- **Surgical biopsy** is the use of surgery to remove some or all of a lump or abnormal area. There are two types of surgical biopsies:
  - **Incisional biopsy** removes part of a lump or a sample of tissue.
  - **Excisional biopsy** removes an entire lump. A small amount of healthy tissue around the lump may also be removed.

You will usually have a biopsy as an outpatient, meaning you will likely go home the same day as the procedure. Whether you will need anesthesia and how long it will take you to recover from a breast biopsy depend on whether it involves surgery. Nonsurgical biopsies (fine-needle or core biopsies) usually do not require anesthesia. Surgical biopsies are done with local or general anesthesia. Recovery takes longer after a surgical biopsy. Talk with your doctor about what to expect during and after your biopsy.

The pathologist will study the biopsy sample and provide the results of their analysis in a pathology report. If the pathologist finds that you have cancer, the pathology report will help you and your doctor understand your cancer and treatment options. The report will include information about:

- where in the breast the cancer started, such as the ducts or lobes
- the tumor grade
- whether the cancer has spread to nearby normal tissue (called invasive breast cancer)

Learn more about the kind of information that can be found in Pathology Reports.

If the biopsy shows breast cancer, the cells will be tested for biomarkers. Learn more about Tests for Breast Cancer Biomarkers.

Tests to stage breast cancer

If you are diagnosed with breast cancer, your doctor will perform more tests to find out if the cancer has spread and if so, how far. Sometimes the cancer is only in the breast. Or, it may have spread from the breast to the lymph nodes or other parts of the body. The process of learning how far the cancer has spread is called staging.

For breast cancer, stages are defined according to many factors, including extent of spread, results of biomarker tests, tumor grade, and, in some cases, multigene tests. It is important to know the stage of breast cancer to plan treatment.

The following tests and procedures may be used find out your breast cancer stage:

Sentinel lymph node biopsy

Breast cancer may spread to nearby lymph nodes, such as those in the underarm area. A sentinel lymph node biopsy can help doctors tell if cancer cells have spread beyond the breast. A sentinel lymph node is the first lymph node to which cancer cells are most likely to spread from the primary tumor. Sometimes, there can be more than one sentinel lymph node.

To locate the sentinel lymph node, a radioactive substance, a blue dye, or both will be injected near the tumor. The surgeon then uses a device to detect lymph nodes that contain the radioactive substance, or they look for lymph nodes that are stained with the blue dye. Once the surgeon locates the sentinel lymph node, they make a small incision in the skin and remove the node.

Sentinel lymph node biopsy of the breast. A radioactive substance and/or blue dye is injected near the tumor (first panel). The injected material is followed visually and/or with a probe that detects radioactivity to find the sentinel nodes (the first lymph nodes to which cancer cells are likely to spread from a primary tumor) (second panel). The sentinel nodes are removed and checked for cancer cells (third panel). A sentinel lymph node biopsy is usually done at the same time the primary tumor is removed, but it can also be done before or after the tumor is removed.

Credit: © Terese Winslow

Enlarge Image

A pathologist then checks the sentinel node for cancer cells. If cancer is found, the surgeon may remove other lymph nodes, either during the same biopsy procedure or during a follow-up surgery. A sentinel lymph node biopsy is an outpatient procedure done under general anesthesia.

Imaging tests to stage breast cancer

The following imaging tests may be done to find out if breast cancer has spread:

A **CT scan** (CAT scan) uses a computer linked to an x-ray machine to make a series of detailed pictures of areas inside the body. The pictures are taken from different angles and are used to create 3-D views of tissues and organs. A dye may be injected into a vein or swallowed to help the organs or tissues show up more clearly. This procedure is also called computed tomography, computerized tomography, or computerized axial tomography. Learn more about Computed Tomography (CT) Scans and Cancer.

A **bone scan** checks if there are rapidly dividing cells, such as cancer cells, in the bone. A very small amount ofradioactive material is injected into a vein and travels through the bloodstream. The radioactive material collects in the areas of the bones with cancer and is detected by a scanner.

A **PET scan** (positron emission tomography scan) uses a small amount of radioactive sugar (also called radioactive glucose) that is injected into a vein. The PET scanner rotates around the body and makes pictures of where sugar is being used by the body. Cancer cells show up brighter in the pictures because they are more active and take up more sugar than normal cells do. When this procedure is done at the same time as a CT scan, it is called a PET-CT scan.

Biomarker tests for breast cancer

All breast cancers are tested for the presence of certain biomarkers, such as hormone receptors (estrogen receptors and progesterone receptors) and HER2. This information is needed to identify the stage of the breast cancer and to plan treatment. Learn more about these tests at Tests for Breast Cancer Biomarkers.

Breast cancer tumor grade

Tumor grade describes how abnormal the cancer cells and tissue look under a microscope and how quickly the cancer cells are likely to grow and spread. The pathologist assigns a score of 1 to 3 to the cells or tissue. A score of 1 means the cells and tumor tissue look the most normal and are less likely to spread. A score of 3 means the cells and tissue look the most abnormal and are more likely to grow fast and spread. Learn more about the breast cancer grading system at Breast Cancer Stages.

Multigene tests to predict risk of recurrence

Multigene tests, also called genomic tests, are lab tests that look at the activity of genes in your breast cancer cells. These tests may help predict the chances that cancer will spread to other parts of the body or come back. And they can help you and your doctor make decisions about treatment.

The tests listed below are often done on a sample of the tumor that was removed during the biopsy or surgery. Most people won't need another procedure for these tests.

There are many types of multigene tests. Examples of tests you might have include:

> Tumor grade for breast cancer is based on how abnormal the cancer cells look under a microscope and how quickly the cancer cells are likely to grow and spread. Grade 1 (well-differentiated) cancer cells look more like normal cells. Grade 2 (moderately differentiated) cancer cells look more abnormal than grade 1 cancer cells but not as abnormal as grade 3 (poorly differentiated) cancer cells.

- **Oncotype DX** looks at the activity of 21 genes in people with early-stage breast cancer that is ER positive, HER2 negative, and has not spread to the lymph nodes or has spread to no more than three lymph nodes. The test helps predict whether breast cancer will spread to other parts of the body. If the risk of the cancer spreading is high, you may have chemotherapy to lower that risk.

The higher the grade number, the more the cancer cells look like abnormal cells and the more likely they will grow and spread quickly.

Credit: © Terese Winslow

Enlarge Image

- **MammaPrint** looks at the activity of 70 genes in the breast cancer tissue of people with early-stage breast cancer that has not spread to the lymph nodes or has spread to no more than three lymph nodes. The test helps predict whether breast cancer will spread to other parts of the body or come back after treatment. If the test shows that the risk of the cancer spreading or returning is high, you may have chemotherapy to lower that risk.
- **Breast Cancer Index** looks at the activity of 11 genes in the breast cancer tissue of people with early-stage, HR-positive breast cancer that has not spread to the lymph nodes or has spread to no more than three lymph nodes. The test helps predict the risk that your breast cancer will come back within 5 to 10 years after diagnosis. If the test shows that the risk of the cancer coming back is high, you may receive 5 more years of hormone therapy to help lower that risk.

Other types of multigene tests are available. Ask your doctor if multigene tests might be right for you and how the results might affect your treatment plan.

Genetic counseling and testing

Your doctor might suggest genetic testing as part of diagnosing your breast cancer. Genetic testing can be done on a sample of your blood, saliva, or from a swab of inside your cheek. Genetic test results show if you were born with a change in the *BRCA1*, *BRCA2*, or another breast cancer risk gene. Knowing if you have a specific genetic change may help your doctor suggest the best treatment for your cancer. These gene changes are sometimes called mutations or pathogenic variants.

Genetic counseling before genetic testing can help you understand your chances of having a gene change that caused your breast cancer and whether genetic testing is needed. For example, genetic testing might be advised if you:

- have a known family history of a change in *BRCA1*, *BRCA2*, or another breast cancer risk gene
- were diagnosed with breast cancer before age 50
- were diagnosed with two or more primary breast cancers (in the same breast or in both breasts)
- have triple-negative breast cancer
- have lobular breast cancer with a family history of diffuse gastric cancer
- have a family history of breast cancer or certain other cancers, such as pancreatic cancer, prostate cancer, or ovarian cancer
- have male breast cancer
- are of Ashkenazi Jewish descent

Genetic counselors can also help you cope with your genetic testing results, including how to discuss the results with family members. They can advise you about whether other members of your family should receive genetic testing to find out if they are at risk of breast cancer or other cancers. Learn more about *BRCA* gene changes and cancer risk and genetic testing for inherited cancer risk.

Waiting for test results

Waiting for cancer test results can be stressful. Ask your doctor when the results will be ready and how you will be notified so you know how long it will be before you hear back. If you don't hear back within the expected time, it's okay to follow up with your doctor.

Sometimes test results are released to your online patient portal before you've met with your doctor. Although it can be difficult to wait to look at the test results until you are able to review them with your doctor, doing so can help avoid confusion about what the results mean.

If you need help dealing with the stress of waiting for test results, reach out to your care team. A social worker can offer support and resources to help you. Whatever the results, your doctor can help you understand next steps.

Getting a second opinion

You may want a second opinion to confirm your breast cancer diagnosis and treatment plan. If you seek a second opinion, you will need to get medical test results and reports from the first doctor to share with the second doctor. The second doctor will review the pathology report, slides, and scans before giving advice. The doctor who gives the second opinion may agree with your first doctor, suggest changes or another approach, or provide more information about your cancer.

To learn more about choosing a doctor and getting a second opinion, visit Finding Cancer Care. You can contact NCI's Cancer Information Service via chat, email, or phone (both in English and Spanish) for help finding a doctor or hospital or getting a second opinion. For questions you might want to ask at your doctor visits, visit Questions to Ask Your Doctor About Cancer.

NIH NATIONAL CANCER INSTITUTE

Stages of Breast Cancer

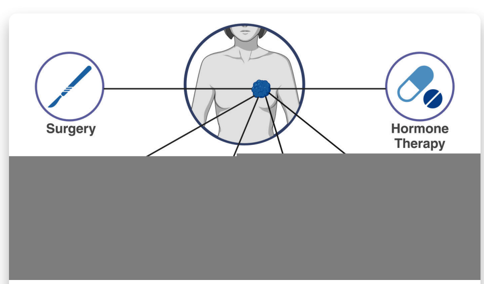
A cancer stage describes the extent of cancer in the body and whether the cancer has spread from where it first formed to other parts of the body. It is important to know the stage of breast cancer to plan your treatment and understand your prognosis.

In breast cancer, stage is based on the size and location of the primary tumor, the spread of cancer to nearby lymph nodes or other parts of the body, the grade of the tumor, and whether certain biomarkers are present.

Learn about tests to stage breast cancer at How Breast Cancer Is Diagnosed in the section, “Tests to stage breast cancer.”

On This Page

How breast cancer stage is determined



There are two types of staging for breast cancer. Staging done before surgery is called clinical prognostic staging, and staging done after surgery is called pathological prognostic staging.

- **Clinical prognostic stage** is used first to assign a stage for anyone diagnosed with breast cancer. It is based on health history, physical examination,

imaging tests (including mammography or ultrasound examination of the lymph nodes), and biopsy findings.
- **Pathological prognostic stage** is used only for people who have surgery as their first treatment for breast cancer. The pathological prognostic stage is based on all clinical information, biomarker status, and laboratory test results from breast tissue and lymph nodes removed during surgery.

Breast Cancer Treatment by Stage

Learn about treatment options for breast cancer by stage.

In both types of staging, a number between 0 and IV (4) is assigned. This number is determined by a combination of three factors: TNM (tumor node metastasis) value, grade, and biomarker status (whether the tumor has certain hormone receptors or high levels of HER2). Results from multigene tests may be used to help stage some breast cancers. Generally, the lower the stage number, the less the cancer has spread.

TNM staging

TNM staging uses T, N, and M categories to describe the size of the primary tumor and whether the cancer has spread to nearby lymph nodes or other parts of the body.

- **Tumor (T):** The size and location of the tumor.
- **Lymph node (N):** The location of the lymph nodes where cancer has spread and the size of the cancer in the lymph nodes.
- **Metastasis (M):** The spread of cancer to other parts of the body.

In breast cancer, stage is based on the TNM status (the size and location of the primary tumor and the spread of cancer to nearby lymph nodes or other parts of the body), the presence of certain biomarkers, and the grade of the tumor.

Credit: © Terese Winslow

Enlarge Image

The numbers or letters after T, N, and M provide more details about the cancer. Higher numbers generally mean the cancer is more advanced.

To learn more, review the full TNM staging descriptions for breast cancer.

The grading system

Tumor grade describes how abnormal the cancer cells and tissue look under a microscope and how quickly the cancer cells are likely to grow and spread. Low-grade cancer cells look more like normal cells and tend to grow and spread more slowly than high-grade cancer cells. To describe how abnormal the cancer cells and tissue are, the pathologist will assess the following three features:

- how much of the tumor tissue has normal breast ducts
- the size and shape of the nuclei in the tumor cells
- how many dividing cells are present, which is a measure of how fast the tumor cells are growing and dividing

For each feature, the pathologist assigns a score of 1 to 3. A score of 1 means the cells and tumor tissue look the most like normal cells and tissue, and a score of 3 means the cells and tissue look the most abnormal. The scores for each feature are added together to get a total score between 3 and 9.

Three grades are possible:

- total score of 3 to 5: G1 (low grade or well differentiated)
- total score of 6 to 7: G2 (intermediate grade or moderately differentiated)
- total score of 8 to 9: G3 (high grade or poorly differentiated)

Biomarker testing

Biomarker testing is used to find out whether breast cancer cells have certain receptors (biomarkers), including:

- **Estrogen receptor (ER).** If the breast cancer cells have estrogen receptors, the cancer cells are called ER positive (ER+). If the breast cancer cells do not have estrogen receptors, the cancer cells are called ER negative (ER-).
- **Progesterone receptor (PR).** If the breast cancer cells have progesterone receptors, the cancer cells are called PR positive (PR+). If the breast cancer cells do not have progesterone receptors, the cancer cells are called PR negative (PR-).
- **Human epidermal growth factor type 2 receptor (HER2/neu or HER2).** If the breast cancer cells have more than the normal amount of HER2 receptors on their surface, the cancer cells are called HER2 positive (HER2+). If the breast cancer cells have a normal amount of HER2 on their surface, the cancer cells are called HER2 negative (HER2-). HER2+ breast cancer is more likely to grow and divide faster than HER2- breast cancer.

Learn more about Tests for Breast Cancer Biomarkers.

Stage 0 breast cancer

Stage 0 is a noninvasive breast cancer, such as ductal carcinoma in situ (DCIS). In stage 0, abnormal cells are found in the breast, but there is no evidence that these cells have spread from where they first formed. Stage 0 breast cancers can be any grade and have any biomarker (HER2, ER, PR) status.

Learn about treatment for DCIS at Ductal Carcinoma in Situ in the section, "How is DCIS treated?"

Stage I (1) breast cancer

In stage I breast cancer, the cancer is small and only in the breast tissue, or it might be found in lymph nodes close to the breast. Stage I can be divided into stage IA and stage IB. The difference is determined by the size of the tumor and whether there are any lymph nodes with cancer.

Learn about treatment of stage I breast cancer.

Stage IA

Stage IA breast cancer generally has a TNM score of T0 or T1, N0 or N1, and M0. It can be any grade and different combinations of HER2, ER, and PR status. To learn more about the different staging factors, visit TNM Staging for Breast Cancer.

TNM	Grade	HER2 Status	ER Status	PR Status
T1, N0, M0	G1	Positive	Positive	Any
T0, N1mi, M0	G1	Positive	Negative	Any
T1, N1mi, M0	G1 or G2	Negative	Positive	Any
T1, N1mi, M0	G1 or G2	Negative	Negative	Positive
T1, N1mi, M0	G2 or G3	Positive	Any	Any
T1, N1mi, M0	G3	Negative	Positive	Positive

Abbreviations: T = tumor N = lymph node M = metastasis, HER2 = human epidermal growth factor type 2 receptor, ER = estrogen receptor, PR = progesterone receptor.

Stage IB

Stage IB breast cancer generally has a TNM score of T0 or T1, N0 or N1, and M0. It can be any grade and different combinations of HER2, ER, and PR status. To learn more about the different staging factors, visit TNM Staging for Breast Cancer.

TNM	Grade	HER2 Status	ER Status	PR Status
T1, N1mi, M0	G1 or G2	Negative	Negative	Negative
T0, N1, M0 T1, N1, M0 T2, N0, M0	G1 or G2	Any	Positive	Positive
T0, N1, M0 T1, N1, M0 T2, N0, M0	G3	Positive	Positive	Positive
T2, N1, M0 T3, N0, M0	Any	Positive	Positive	Positive
T1, N1mi, M0	G3	Negative	Positive	Negative
T1, N1mi, M0	G3	Negative	Negative	Any
T0, N1, M0 T1, N1, M0 T2, N0, M0	G3	Positive	Positive	Positive

Stage II (2) breast cancer

In stage II breast cancer, there is cancer in the breast or nearby lymph nodes or both. This stage is divided into groups: Stage IIA and Stage IIB. The difference is determined by the size of the tumor and whether the breast cancer has spread to the lymph nodes.

Learn about treatment of stage II breast cancer.

Stage IIA

Stage IIA breast cancer generally has a TNM score of T0, T1, or T2; N0 or N1; and M0. It can be any grade and different combinations of HER2, ER, and PR status. To learn more about the different staging factors, visit TNM Staging for Breast Cancer.

TNM	Grade	HER2 Status	ER Status	PR Status
T0, N1, M0 T1, N1, M0 T2, N0, M0	G1 or G2	Positive	Positive	Negative
T0, N1, M0 T1, N1, M0 T2, N0, M0	G1 or G2	Positive	Negative	Any
T0, N1, M0 T1, N1, M0 T2, N0, M0	G1 or G2	Negative	Positive	Negative
T0, N1, M0 T1, N1, M0 T2, N0, M0	G1	Negative	Negative	Any
T0, N1, M0 T1, N1, M0 T2, N0, M0	G2	Negative	Negative	Positive

Stage IIB

Stage IIB breast cancer generally has a TNM score of T0, T1, or T2; N0 or N1; and M0. It can be any grade and different combinations of HER2, ER, and PR status. To learn more about the different staging factors, visit TNM Staging for Breast Cancer.

TNM	Grade	HER2 Status	ER Status	PR Status
T0, N1, M0 T1, N1, M0 T2, N0, M0	G2	Negative	Negative	Negative

TNM	Grade	HER2 Status	ER Status	PR Status
T0, N1, M0 T1, N1, M0 T2, N0, M0	G3	Negative	Positive	Negative
T0, N1, M0 T1, N1, M0 T2, N0, M0	G3	Negative	Negative	Any
T2, N1, M0 T3, N0, M0	G1 or G2	Positive	Negative	Negative
T2, N1, M0 T3, N0, M0	G1 or G2	Negative	Positive	Negative
T2, N1, M0 T3, N0, M0	G1 or G2	Negative	Negative	Any
T2, N1, M0 T3, N0, M0	G3	Positive	Positive	Negative
T2, N1, M0 T3, N0, M0	G3	Positive	Negative	Any
T2, N1, M0 T3, N0, M0	G3	Negative	Positive	Positive
T0, N2, M0 T1, N2, M0 T2, N2, M0 T3, N1, M0 T3, N2, M0	G3	Positive	Positive	Positive

Stage III (3) breast cancer

In stage III breast cancer, the cancer is found in the lymph nodes close to the breast, the skin of the breast, or the chest wall.

[Learn about treatment of stage III breast cancer.]

Stage IIIA

Stage IIIA breast cancer generally has a TNM score of any T, any N, and M0. It can be any grade and different combinations of HER2, ER, and PR status. To learn more about the different staging factors, visit TNM Staging for Breast Cancer.

TNM	Grade	HER2 Status	ER Status	PR Status
T2, N1, M0 T3, N0, M0	G3	Negative	Positive	Negative
T2, N1, M0 T3, N0, M0	G3	Negative	Negative	Positive
T0, N2, M0 T1, N2, M0 T2, N2, M0 T3, N1, M0 T3, N2, M0	Any	Positive	Positive	Negative
T0, N2, M0 T1, N2, M0 T2, N2, M0 T3, N1, M0 T3, N2, M0	Any	Positive	Negative	Any
T0, N2, M0 T1, N2, M0 T2, N2, M0 T3, N1, M0 T3, N2, M0	G1 or G2	Negative	Positive	Negative
T0, N2, M0 T1, N2, M0 T2, N2, M0 T3, N1, M0 T3, N2, M0	G2	Negative	Negative	Positive
T0, N2, M0 T1, N2, M0	G3	Negative	Positive	Positive

TNM	Grade	HER2 Status	ER Status	PR Status
T2, N2, M0 T3, N1, M0 T3, N2, M0				
T4, N0, M0 T4, N1, M0 T4, N2, M0 Any T, N3, M0	G1 or G2	Positive	Positive	Positive

Stage IIIB

Stage IIIB breast cancer generally has a TNM score of any T, any N, and M0. It can be any grade and different combinations of HER2, ER, and PR status. To learn more about the different staging factors, visit TNM Staging for Breast Cancer.

TNM	Grade	HER2 Status	ER Status	PR Status
T2, N1, M0 T3, N0, M0	G2 or G3	Negative	Negative	Negative
T0, N2, M0 T1, N2, M0 T2, N2, M0 T3, N1, M0 T3, N2, M0	G1 or G2	Negative	Negative	Negative
T0, N2, M0 T1, N2, M0 T2, N2, M0 T3, N1, M0 T3, N2, M0	G3	Negative	Positive	Negative
T0, N2, M0 T1, N2, M0 T2, N2, M0 T3, N1, M0 T3, N2, M0	G3	Negative	Negative	Positive

TNM	Grade	HER2 Status	ER Status	PR Status
T4, N0, M0 T4, N1, M0 T4, N2, M0 Any T, N3, M0	G1 or G2	Positive	Positive	Negative
T4, N0, M0 T4, N1, M0 T4, N2, M0 Any T, N3, M0	G1 or G2	Positive	Negative	Any
T4, N0, M0 T4, N1, M0 T4, N2, M0 Any T, N3, M0	G1 or G2	Negative	Positive	Any
T4, N0, M0 T4, N1, M0 T4, N2, M0 Any T, N3, M0	G1 or G2	Negative	Negative	Positive
T4, N0, M0 T4, N1, M0 T4, N2, M0 Any T, N3, M0	G3	Positive	Any	Any
T4, N0, M0 T4, N1, M0 T4, N2, M0 Any T, N3, M0	G3	Negative	Positive	Positive

Stage IIIC

Stage IIIC breast cancer generally has a TNM score of any T, any N, and M0. It can be any grade and different combinations of HER2, ER, and PR status. To learn more about the different staging factors, visit TNM Staging for Breast Cancer.

TNM	Grade	HER2 Status	ER Status	PR Status
T0, N2, M0 T1, N2, M0 T2, N2, M0 T3, N1, M0 T3, N2, M0	G3	Negative	Negative	Negative
T4, N0, M0 T4, N1, M0 T4, N2, M0 Any T, N3, M0	G1 or G2	Negative	Negative	Negative
T4, N0, M0 T4, N1, M0 T4, N2, M0 Any T, N3, M0	G3	Negative	Positive	Negative
T4, N0, M0 T4, N1, M0 T4, N2, M0 Any T, N3, M0	G3	Negative	Negative	Any

Stage IV (4) or metastatic breast cancer

Stage IV breast cancer means that the cancer has spread to other parts of the body, such as the bones, liver, or lungs. This stage is also called metastatic breast cancer.

Stage IV breast cancer generally has a TNM score of any T, any N, and M1. It can be any grade and different combinations of HER2, ER, and PR status. To learn more about the different staging factors, visit TNM Staging for Breast Cancer.

TNM	Grade	HER2 Status	ER Status	PR Status
Any T, Any N, M1	Any	Any	Any	Any

Learn about treatment for metastatic breast cancer at Metastatic Breast Cancer in the section, “How is metastatic breast cancer treated?”

Recurrent breast cancer

Recurrent breast cancer is cancer that has recurred (come back) after it has been treated. Breast cancer may come back in the breast (local recurrence); nearby lymph nodes, chest, or skin (regional recurrence); or other parts of the body, such as the liver, lung, or bone (distant or metastatic recurrence).

To figure out the type of recurrence you have, you will have many of the same tests you had when your cancer was first diagnosed, such as lab tests and imaging procedures. These tests help determine where in your body the cancer has returned, if it has spread, and how far. Your doctor may refer to this new assessment of your cancer as “restaging.”

After these tests, the doctor may assign a new stage to the cancer. A letter “r” will be added to the beginning of the new stage to reflect the restaging.

Learn about how recurrent breast cancer is treated at Breast Cancer Recurrence in the section, “How is recurrent breast cancer treated?”

Breast Cancer Treatment

Different types of treatment are available for breast cancer. You and your cancer care team will work together to decide your treatment plan, which may include more than one type of treatment.

Your treatment options may include both local and systemic treatments. **Local treatments**, such as surgery and radiation, are directed at the area with cancer and not the whole body. **Systemic treatments**, such as chemotherapy, are drugs that can reach cancer cells throughout the body.

There are different treatment approaches for each type and stage of breast cancer. To learn more about treatment for specific types and stages of breast cancer, review **What kind of treatment will I receive?** below.

Consulte Tratamiento del cáncer de mama (seno) para obtener información sobre el tratamiento del cáncer de mama en español.

Join a Clinical Trial

Joining a clinical trial may be an option. There are different types of clinical trials for people with breast cancer. You can use the clinical trial search to find NCI-supported cancer clinical trials that are accepting participants.

You can also review a list of all current NCI-supported Breast Cancer Clinical Trials. For help finding a clinical trial, contact NCI's Cancer Information Service.

Local treatments

Surgery for Breast Cancer

Learn about surgery to treat breast cancer and access more information about lumpectomy and mastectomy, the two main types of surgery used.

Radiation for Breast Cancer

You may receive radiation therapy as part of your breast cancer treatment. Learn about the types of radiation therapy and possible side effects.

Systemic treatments

Chemotherapy for Breast Cancer

Not everyone with breast cancer will get chemotherapy. Learn about who gets chemo and why and find a list of drugs that may be prescribed.

Hormone Therapy for Breast Cancer

Hormone therapy helps lower the risk of recurrence of certain types of breast cancer. Learn about types of hormone therapy, common side effects, and more.

Immunotherapy for Breast Cancer

Immunotherapy may be used to treat triple-negative breast cancer. Learn about immunotherapy, when it is given, and common side effects.

Targeted Therapy for Breast Cancer

Targeted therapy is a treatment for certain types of breast cancer. Learn about types of targeted therapy, common side effects, and more.

What kind of treatment will I receive?

Breast Cancer Treatment by Stage

The stage of breast cancer is an important factor in determining which type of treatment may work best for you. Learn more about treatment options for breast cancer by stage.

Breast Cancer During Pregnancy

Learn about breast cancer during pregnancy, including challenges in detection and options for treatment in women who are pregnant or breastfeeding.

Ductal Carcinoma in Situ (DCIS)

DCIS is diagnosed when abnormal cells are found in the milk ducts of the breast. Learn more about DCIS, including treatment options and prognosis.

Inflammatory Breast Cancer Treatment

Inflammatory breast cancer treatments may include surgery, radiation therapy, chemotherapy, and more. Learn about the different ways inflammatory breast cancer can be treated.

Triple-Negative Breast Cancer Treatment

Triple-negative breast cancer treatments may include surgery, chemotherapy, and more. Learn about the different ways triple-negative breast cancer can be treated.

Paget Disease of the Breast

Paget disease of the breast involves the nipple and areola. Learn about symptoms, diagnosis, and treatment of this rare type of breast cancer.

Breast Cancer Prognosis and Survival Rates

If you or someone close to you has just been diagnosed with breast cancer you may be asking: How serious is this cancer? What is the prognosis? What are survival rates among people with similar diagnoses?

Prognosis and survival rates vary from person to person and depend on many factors. Talk with your doctor to get a better understanding of the expected outcome of your cancer.

The survival statistics on this page refer to women with breast cancer. For information about prognosis and survival rates for men, visit Male Breast Cancer.

On This Page

Understanding your breast cancer prognosis

A breast cancer prognosis is the likely outcome or course of breast cancer. It may be helpful to think of a prognosis as an estimate or prediction. Understanding your prognosis will help you talk with your doctor and make decisions about treatment.

Many factors affect your breast cancer prognosis, including:

- **Type of breast cancer**, such as ductal carcinoma in situ, invasive ductal carcinoma, inflammatory breast cancer, and triple-negative breast cancer. Learn about types of breast cancer.
- **The size of the tumor and whether it has spread beyond the breast tissue** to the lymph nodes or to other places in the body.
- **Tumor grade**, which describes how abnormal the cells look under the microscope, as well as how fast they are likely to grow and spread.
- **Biomarker status**, in particular whether your tumor has receptors for the hormones estrogen and/or progesterone or has high levels of a protein called HER2. Learn about

breast cancer biomarkers.

- **Personal factors**, such as your age, medical history, family history, menopausal status, and ethnicity.
- **Treatment response**, in particular how your cancer responds to treatment given before surgery.
- **Results from gene expression tests** such as OncotypeDX, Mammaprint, or Breast Cancer Index. These tests, which are also called gene expression profiling or multigene tests, help predict the chances that cancer will spread or come back after surgery.

Your doctor may tell you that you have a certain prognostic stage—stage 1, stage 2, stage 3, or stage 4. The prognostic stage combines information about several of the prognostic factors listed above, including the size and spread of the tumor, tumor grade, whether it has estrogen and/or progesterone receptors and HER2, and its score on a gene expression test. Learn more about Breast Cancer Stages.

As part of a discussion of breast cancer prognosis, your doctor can tell you about the chance that your cancer will return after treatment and how different treatments might affect that chance. To learn more about breast cancer recurrence and how it is diagnosed and treated, visit Recurrent Breast Cancer.

Survival statistics for breast cancer

Doctors estimate breast cancer prognosis by using statistics collected over many years from people with breast cancer. One commonly used statistic is the 5-year relative survival rate. This statistic tells you what percentage of people with the same type and stage of breast cancer are alive 5 years after their cancer was diagnosed, compared with people in the overall population.

The most recent 5-year relative survival rates for female breast cancer from NCI's Surveillance, Epidemiology, and End Results (SEER) program, representing the period 2013–2019, are as follows:

- For all types of breast cancer combined, 90.8%
- For all types of breast cancer, based on how far breast cancer has spread at the time of diagnosis,
    - 99.3% for localized breast cancer (cancer is found only in the breast tissue and has not spread to nearby lymph nodes or other parts of the body)

    - 86.3% for regional breast cancer (cancer has spread beyond the breast tissue to nearby lymph nodes or organs)
    - 31% for distant breast cancer (cancer has spread, or metastasized, from the original tumor to distant locations in the body; this is sometimes referred to as stage 4 or metastatic breast cancer)
- For subtypes of breast cancer, based on hormone receptor and HER2 biomarkers,
    - 94.8% for HR+/HER2- breast cancer (localized, 100.0%; regional, 90.2%; distant, 34.0%)
    - 91.0% for HR+/HER2+ breast cancer (localized, 99.1%; regional, 89.8%; distant, 45.6%)
    - 85.6% for HR-/HER2+ breast cancer (localized, 97.2%; regional, 84.0%; distant, 39.5%)
    - 77.6% for HR-/HER2- (triple-negative) breast cancer (localized, 91.8%; regional, 66.2%; distant, 12.8%)

Talking with your doctor to make sense of cancer statistics

Because survival statistics are based on large groups of people, they cannot be used to predict exactly what will happen to you. Your doctor can help you understand how these statistics relate to your own situation.

When reviewing survival statistics, it’s important to keep in mind that:

- responses to cancer treatment vary, even among women with the same type of breast cancer who are receiving the same treatment.
- it takes several years to see the effect of newer and better treatments, so the positive impact of improved treatments may not be reflected in current statistics.

Learn more about a cancer prognosis and watch videos of people with cancer talking with their doctor in Understanding Cancer Prognosis.

Additional breast cancer statistics

NCI’s SEER program provides additional breast cancer statistics, including statistics on incidence, lifetime risk, prevalence, and deaths. Statistics are available for different racial and ethnic groups and different age groups.

Living with Breast Cancer & Survivorship

You may have just learned that you have breast cancer. Or you may be in treatment, finishing treatment, or have a friend or family member with cancer. Wherever you are in dealing with cancer, from the time of diagnosis through the rest of your life, you are thought of as a survivor. As a result, what being a breast cancer survivor means to you may change over time. Or you may not feel that the term applies to you, and that is okay, too.

Some survivors say they would not have been able to cope without the help and love of their family members.

Credit: iStock

Having cancer changes your life and the lives of those around you. The symptoms and side effects of the disease and its treatment may cause certain physical changes, but they can also affect the way you feel and how you live.

Learn more about the unique challenges cancer survivors face and find resources to help you cope at Cancer Survivorship.

Coping with breast cancer treatment

Cancer treatment can cause side effects or changes to your body and how you feel. You may have different side effects even when you are given the same treatment for the same type of cancer as other people. It's normal to feel uncertain or fearful about side effects that you may have. Keep in mind your health care team can help prevent and relieve any side effects.

Side effects of cancer treatment

Breast cancer treatments can cause many physical side effects that affect healthy tissues or organs. The type of side effects you may have depends on the type of treatments you have and their doses. Some side effects that many people with breast cancer have include lymphedema and fertility issues.

Learn more about the possible side effects of cancer treatment at Side Effects of Cancer Treatment.

Emotional effects of cancer treatment

Breast cancer and its treatment can be overwhelming and cause many emotional effects. Just as cancer affects your physical health, it can bring up a wide range of emotions you're not used to dealing with. It can also make existing feelings seem more intense.

Learn more about how to understand and cope with new, and sometimes difficult, thoughts and feelings at Emotions and Cancer.

Changes in body image and sexuality

Breast surgery, deciding whether to have breast reconstruction, and hair loss are some examples of how cancer treatment may affect how you look and feel about yourself. Other treatments for breast cancer also may affect your body, how others see you, and how you see yourself. Coping with these changes to your body can be hard, but there are resources that can help you.

Learn more about coping with changes to your body in How Cancer Affects Your Self-Image and Sexuality.

Follow-up care

Many breast cancer survivors need to visit their doctor regularly to get follow-up exams or tests. Planning and scheduling these appointments can be stressful and time-consuming, but making regular visits with your health care team can help you feel supported.

Your doctor may give you a survivorship care plan that details the types of cancer treatment you received and a plan for your follow-up care. At follow-up appointments, your doctor may do imaging and other tests to check for a recurrence, such as:

- physical exam
- clinical breast exam
- mammogram

Getting Financial Support

Even if you have health insurance, you may face financial challenges and need help dealing with the costs of breast cancer treatment. For tips and ways to cope, visit

- breast MRI
- neurological exam
- pelvic exam

Learn more about the health care you should be receiving after treatment at Follow-Up Medical Care.

Managing Cancer Costs and Medical Information.

Coping with fear of breast cancer recurrence

You may feel a mix of relief and worry when treatment ends. When you stop having a treatment that was working, it can cause fear that cancer could come back. Tests and scans that are part of normal follow-up care can create anxiety called “scanxiety.” Physical changes in your body also can cause worry. You may fear that any pain or discomfort may be due to cancer’s return. Although these feelings are uncomfortable, they can improve with time and help.

Learn more about how to cope with fear of recurrence at Life After Cancer Treatment.

Late effects of cancer treatment

Late effects are problems caused by cancer treatment that may not show up for months or years after treatment. These problems are specific to certain types of treatments and the amount received. Like side effects that you may have during treatment, late effects can be very different from person to person.

Learn about late effects of treatment and how to cope at Late Effects of Cancer Treatment.

Getting support

No matter where you are in your cancer journey, you or your caregivers may need help coping. Know that you are not alone. Seek support from family and friends, your health care team, and support groups.

Get advice on how to ask for help in Adjusting to Cancer, or learn more about Cancer Support Groups.

Support for caregivers

If you are caring for someone who has cancer, it's important to know what resources are available to you. Learn more in Support for Caregivers of Cancer Patients.

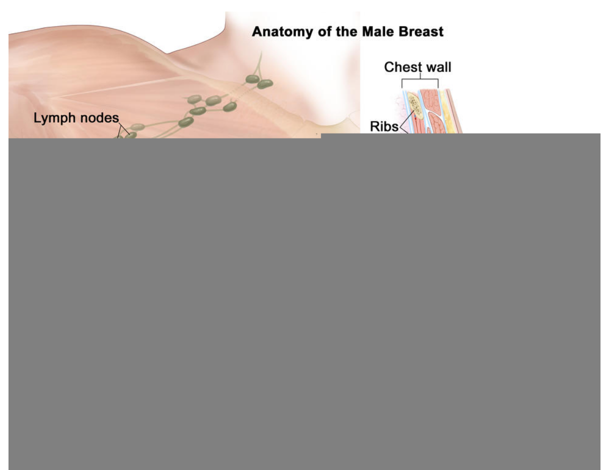
NATIONAL CANCER INSTITUTE

Male Breast Cancer

On This Page

What is male breast cancer?

Male breast cancer is breast cancer that develops in men. Breast cancer is much more common in women, but men can get breast cancer because they also have breast tissue where cancer can start. Fewer than 1 in 100 breast cancers in the United States occur in men. Men of any age can get breast cancer, but it usually occurs at a later age in men than in women.



Anatomy of the male breast. The nipple and areola are shown on the outside of the breast. The lymph nodes, fatty tissue, ducts, and other parts of the inside of the breast are also shown.

Credit: © Terese Winslow

Enlarge Image

Most male breast cancers start in the milk ducts of the breast and are called ductal cancers. Both female and male breasts have milk ducts, but in male breasts, milk ducts do not develop in the same way as they do in female breasts. Male breasts may also have milk glands called lobules, but far fewer than in female breasts. Cancers that start in the lobules, called lobular cancers, are rare in men.

Other types of breast cancer are extremely rare in men, but they can happen. These include phyllodes tumor, Paget disease of the breast, and inflammatory breast cancer.

When abnormal cells are found in the ducts and have not spread to other tissues in the breast, it is called ductal carcinoma in situ. Invasive cancers have spread into surrounding breast tissue and can spread to nearby lymph nodes or other organs throughout the body. Most breast cancers in men are invasive.

What are risk factors for male breast cancer?

Older age is a risk factor for most cancers, including breast cancer in men. In addition, if you are male, you may have an increased risk of breast cancer if you have:

- a history of radiation therapy directed at the chest
- one or more female relatives who have had breast cancer
- inherited changes in the *BRCA1* or *BRCA2* genes or in other genes that increase breast cancer risk
- a condition linked to higher-than-normal estrogen levels in the body:
    - Klinefelter syndrome
    - liver disease (cirrhosis)
    - testicular problems, including inflamed testicles (orchitis), an undescended testicle, or surgery to remove one or both testicles
    - obesity

Screening for male breast cancer

If you think you may have risk factors for male breast cancer, talk to your doctor. Screening mammograms are not usually recommended for men, even those who are at increased risk of breast cancer. But your doctor may suggest staying alert to any changes in your breast or the skin of your breast and getting regular clinical breast exams.

What are the symptoms of breast cancer in men?

Signs and symptoms of breast cancer in men include:

- a lump or thickening in or near the breast (most common)
- a nipple that changes in shape or the direction it is pointing
- a nipple that is inverted

- fluid from the nipple that may be clear or bloody
- an open wound (ulcer) in the skin of the breast
- scaly, red, or swollen skin on the breast, nipple, or areola
- dimpling or ridges on the skin that resemble an orange peel
- swollen lymph nodes under the arm or near the collarbone

Many breast changes in men are signs of less serious or benign conditions, but it is important to check with your doctor if you notice any unusual breast changes.

How is breast cancer diagnosed in men?

If you are male and have symptoms of breast cancer, your doctor will need to find out if they are due to cancer or another condition. Your doctor may:

- do a physical exam, including a clinical breast exam
- ask about your personal and family medical history to learn more about your symptoms and risk factors for breast cancer
- do imaging tests, such as a breast ultrasound and mammogram
- do a breast biopsy, usually a core biopsy

If it is cancer, additional tests may include:

- **Biomarker tests** check the cancer cells for hormone receptors and HER2 protein that may help plan treatment. Male breast cancer is almost always hormone receptor positive. Learn more about Tests for Breast Cancer Biomarkers.
- **Bone scan** checks to see if cancer has spread to your bones.
- **PET scan** or **CT scan** are imaging tests that help determine the extent of spread of the cancer.
- **Genetic testing** looks for inherited mutations in *BRCA1*, *BRCA2*, and several other genes that increase the risk of breast cancer. Such mutations, if present, can affect treatment decisions. They also have implications for family members.

Your doctor will assign a stage to the cancer based on a combination of the extent of spread, hormone receptor and HER2 status, and other tumor features, such as tumor grade.

Learn more about the different stages of breast cancer and tests used to diagnose it at Breast Cancer Stages and How Breast Cancer Is Diagnosed.

How is male breast cancer treated?

Male breast cancer is usually treated in the same way as breast cancer in women. However, men are more likely than women to have a mastectomy (surgery to remove the whole breast) rather than a lumpectomy (breast-conserving surgery) to remove the cancer because they have less breast tissue. But some male breast cancers can be treated with a lumpectomy, depending on the size of the cancer, the amount of breast tissue the person has, and other factors.

Learn more about breast cancer treatment at:

- Ductal Carcinoma in Situ
- Breast Cancer Treatment by Stage
- Metastatic Breast Cancer

What is the survival rate and prognosis for men with breast cancer?

Doctors estimate the prognosis for men with breast cancer by using statistics collected over many years. One common statistic is the 5-year relative survival rate. The 5-year relative survival rate tells you what percent of people with the same type and stage of male breast cancer are alive 5 years after their cancer was diagnosed, compared with people in the overall population.

Men with breast cancer have a slightly lower 5-year survival rate than their female peers. This may be because male breast cancer is often diagnosed at a later stage than female breast cancer. Race also affects survival. Black men are usually diagnosed with a later stage of breast cancer than White men and are more likely to die of their disease.

More research is needed to find better treatments for male breast cancer and to better understand differences between male and female breast cancers.

The 5-year relative survival rates for people with male breast cancer are:

- 95% for localized male breast cancer (cancer is in the breast only)
- 84% for regional male breast cancer (cancer has spread beyond the breast to nearby lymph nodes or organs)
- 20% for metastatic male breast cancer (cancer has spread beyond the breast to a distant part of the body)

To learn more about factors that affect breast cancer prognosis, visit Breast Cancer Prognosis and Survival Rates.

NIH NATIONAL CANCER INSTITUTE

Breast Lumps and Breast Cancer in Children

A breast lump is a swelling, bump, or mass in the breasts. Different types of breast lumps can happen in children and adolescents, and most are benign (not cancer). Breast lumps may not cause symptoms or need treatment, but it is still important to contact your child's doctor if you or your child notices any unusual changes.

On This Page

Types of breast lumps in children

The following types of breast lumps can occur in children and adolescents:

- **Fibroadenomas** are the most common benign breast lumps seen in children aged 18 and younger. These lumps usually form in the connective tissue and glandular tissue of the breast. They are usually painless and rarely cause symptoms. In some cases, fibroadenomas may grow very large and need to be removed by surgery. They may also stay the same size or disappear without treatment. Learn more about Benign and Precancerous Breast Lumps and Conditions.
- **Phyllodes tumors** are rare tumors that usually form in the connective tissue of the breast. Phyllodes tumors tend to grow quickly and get large, but they rarely spread to other parts of the body. Most are benign, but some may be malignant (cancer) or borderline (in between benign and malignant). Phyllodes tumors are usually removed by surgery, but they can come back. They are most common in women between 40 and 50 years of age but can also develop in children and adolescents. Learn more about Benign and Precancerous Breast Lumps and Conditions.

- **Fibrocystic breast changes** are common and benign changes within the breast that can make the breasts feel bumpy or irregular in consistency. There may also be breast swelling or discomfort, sensitive nipples, and itching. These changes are likely due to changes in hormone levels during the menstrual cycle. Fibrocystic breast changes can occur at any age but are most common in younger women. They don't require treatment. Learn more about Benign and Precancerous Breast Lumps and Conditions.
- **Breast cancer** is possible in children and adolescents but very rare. Less than 5% of all breast cancers occur in females younger than 40. Learn more about breast cancer, including symptoms, risk factors, diagnosis, and treatment at What Is Breast Cancer?

Causes and risk factors for breast lumps in children

Often, the exact cause of benign breast lumps is unknown.

Children and adolescents who have any of the following syndromes have an increased risk of fibroadenoma:

- Beckwith-Wiedemann syndrome
- Maffucci syndrome
- Cowden syndrome

Children and adolescents who have Li-Fraumeni syndrome have an increased risk of phyllodes tumor.

Symptoms of breast lumps in children

Symptoms of a breast lump may not occur until the lump has grown bigger. Symptoms of benign breast lumps may include:

- a lump in the breast, underarm, or around the nipple
- a change in the size or shape of the breast that is not associated with puberty
- breast pain or discomfort

Check with your child's doctor if your child has any of these symptoms.

Tests to diagnose breast lumps in children

If your child has symptoms of a breast lump, their doctor will need to find out the cause. The doctor will ask about your child's personal and family medical history and do a physical exam. Depending on these results, they may recommend other tests or procedures to help make a diagnosis. The results of the tests will also help you and your child's doctor plan treatment.

The following tests may be used to diagnose breast lumps in children and adolescents:

- A **clinical breast exam** is an exam of the breast by a doctor or other health professional. The doctor will carefully feel the breast and under the arm for lumps or anything else that seems unusual.
- An **ultrasound exam** uses high-energy sound waves (ultrasound), which bounce off internal tissues or organs and make echoes. The echoes form a picture of body tissues called a sonogram.
- **Biopsy** is the removal of a sample of tissue from the breast so that a pathologist can view it under a microscope. Biopsies may be done in kids and teens if the lump is large or fast growing.

Treatment of breast lumps in children

There are different types of treatment for children and adolescents with benign breast lumps. You and your child's care team will work together to decide treatment.

Treatment of breast lumps in children and adolescents may include:

- Watchful waiting, which is closely monitoring a person's condition without giving any treatment until symptoms appear or change. Watchful waiting may be done for fibroadenomas because they may disappear without treatment.
- Surgery to remove fibroadenomas that have not disappeared or phyllodes tumors. The whole breast is not removed during surgery. Surgery may also be used for lumps that come back.

NIH NATIONAL CANCER INSTITUTE

Breast Cancer During Pregnancy

A new breast cancer diagnosis that occurs in a woman who is pregnant, within a year of giving birth, or anytime during lactation is called pregnancy-associated breast cancer. Breast cancer is one of the most common cancers diagnosed during pregnancy. It occurs in about 1 out of every 3,000 pregnancies, most often in women ages 32 to 38 years. The incidence of pregnancy-associated breast cancer is likely to increase because the average age at pregnancy is increasing and overall incidence of breast cancer increases with age.

Women can be treated for breast cancer during pregnancy, but the type of treatment and when it is given may be affected by the pregnancy. Localized breast cancer does not appear to harm the fetus, and breast cancer cells do not seem to pass from the mother to the fetus.

Sometimes breast cancer occurs in women who are pregnant or have just given birth. Talk to your doctor if you notice any changes in your breasts that you do not expect or that worry you.

Credit: iStock

On This Page

- Finding breast cancer during pregnancy
- Diagnosing breast cancer during pregnancy
- Prognosis for women with breast cancer during pregnancy
- Treating breast cancer during pregnancy
- Clinical trials
- Coping with breast cancer during pregnancy

Finding breast cancer during pregnancy

Breast cancer can be difficult to detect during pregnancy. Normal breast changes that occur during pregnancy, such as larger, more tender, and lumpy-feeling breasts, can make it difficult to find small masses during clinical breast exams and self-exams.

Increased breast density during pregnancy can also make it harder to find breast cancer with a mammogram. Breast density is higher during pregnancy because hormonal shifts increase glandular tissue and reduce fat in preparation for lactation. Additionally, women who are pregnant or nursing are often below the age at which mammogram screening is recommended.

Because these factors can delay a breast cancer diagnosis and because younger women are more likely to have aggressive forms of the disease, breast cancer is often found at a later stage in women who are pregnant or nursing.

If you are pregnant or nursing, you should receive clinical breast exams during regular prenatal and postnatal check-ups. However, most women diagnosed with pregnancy-associated breast cancer find a lump on their own. If you are pregnant or nursing, be aware of changes in your breasts. Talk to your doctor right away if you notice any changes that worry you.

Diagnosing breast cancer during pregnancy

The same tests used to diagnose breast cancer in women who are not pregnant can be used in women who are pregnant. The first test to evaluate a suspicious mass is typically an ultrasound, which does not involve radiation. Mammograms involve only a small amount of radiation and can safely be used to further evaluate a mass. Biopsies are also safe to have during pregnancy.

If you are diagnosed with breast cancer, additional tests may be needed to determine the stage of the disease. Some tests, such as CT or PET-CT scans that are used to stage breast cancer in women who aren't pregnant, may be dangerous for a developing fetus or a nursing baby due to radiation or use of dyes or radioactive tracers. These tests are done only if absolutely necessary. Certain precautions, such as covering the abdomen with a lead shield, are taken to ensure that the fetus is exposed to the lowest possible amount of radiation.

Different tests and procedures may be used to stage breast cancer in women who are pregnant or nursing than in women who are not. Tests that may safely be used to stage breast cancer during pregnancy include ultrasound and MRI with limited use of contrast.

To learn more about the tests and procedures used to diagnose and stage breast cancer, visit How Breast Cancer Is Diagnosed.

Prognosis for women with breast cancer during pregnancy

The prognoses for pregnant and nonpregnant women with breast cancer may differ. Research results have been mixed. Some studies show prognosis is similar if cancers are found at the same stage; however, other research shows worse outcomes for those diagnosed when pregnant.

Ending a pregnancy or stopping lactation does not necessarily improve a woman's chance of survival, although this depends on the type of cancer and stage at diagnosis.

For women who have had breast cancer in the past, getting pregnant does not seem to increase the risk that the cancer will come back. However, some doctors recommend waiting 2 years after treatment for breast cancer before getting pregnant so that an early return of the cancer would be more easily detected. For women on adjuvant hormone therapy, subsequent pregnancy attempts require stopping it and then waiting a few months before trying to become pregnant. Hormone therapy can then be started again after the baby is born. To learn more about pregnancy after cancer, visit Female Fertility and Cancer Treatment.

Treating breast cancer during pregnancy

Treatment options for pregnant women depend on the stage of the disease and the trimester of the pregnancy.

Pregnant women with early-stage breast cancer (stage I and stage II) are usually treated in the same way as women who are not pregnant, with some changes in the timing of certain treatments to protect the fetus. Treatment may be more complicated in pregnant women with late-stage breast cancer (stage III and stage IV) because some treatments may need to begin quickly and could harm the fetus. Ending a pregnancy may be considered if the cancer is fast-growing or advanced and treatment is needed right away. The ability to safely terminate a pregnancy because of an advanced cancer diagnosis may vary based on your state of residence.

Treatment for breast cancer during pregnancy may include:

- **Surgery.** Most pregnant women with breast cancer have a lumpectomy, also called breast-conserving surgery, or a modified radical mastectomy to remove the breast. Some of the lymph nodes under the arm may be removed so a pathologist can check them under a microscope for signs of cancer. If surgery is planned and the woman has already given birth, breastfeeding should be stopped to reduce blood flow in the breasts and make them smaller. Learn more about Breast Cancer Surgery, including reconstruction.

- **Chemotherapy.** After the first 3 months of pregnancy, certain types of chemotherapy may be given before or after surgery. Chemotherapy given at this time does not usually harm the fetus but may cause early labor or low birth weight. Chemotherapy also should be avoided 3 to 4 weeks before delivery, because it can cause delivery complications for both the mother and baby. Women receiving chemotherapy should not breastfeed. Many chemotherapy drugs may be found in high levels in breast milk and could harm the nursing baby. Learn more about Chemotherapy for Breast Cancer.
- **Radiation therapy.** Delaying radiation therapy until after the baby is born is usually the safest option. Women with late-stage breast cancer (stage III or stage IV) may consider external radiation therapy after a careful assessment of the risks and benefits. Talk with your doctor about the risks and benefits of radiation therapy as you discuss your treatment plan. Learn more about Radiation for Breast Cancer.
- **Other treatments.** Hormone therapy, targeted therapy, and immunotherapy are usually not given to pregnant or nursing women because these treatments are harmful to a fetus or a nursing baby.

Clinical trials

Joining a clinical trial specifically designed to include women who are pregnant or nursing may be an option. Examples of these types of trials may include a treatment trial that tests new treatments or new ways of using existing treatments in pregnant women or a registry trial that tracks outcomes.

However, despite the need for research, pregnant and lactating women are often excluded from breast cancer clinical trials because of the complexity of managing treatment and concerns about potential harm to the fetus.

You can use the clinical trial search to find NCI-supported cancer clinical trials that are accepting participants. This search allows you to filter trials based on the type of cancer, your age, and where the trials are being done. You can also review a list of all current Breast Cancer Clinical Trials.

Learn more about clinical trials at Cancer Clinical Trial Information for Patients and Caregivers.

Coping with breast cancer during pregnancy

Breast cancer during pregnancy is especially challenging, both mentally and physically. It is important that you bring up any questions and concerns you have with your health care team throughout your care and treatment to get the support you need.

In addition to support from your health care team and family, one-on-one counseling or in-person and online support groups can help you cope and improve your mental wellbeing. To learn more about how to manage the physical and emotional effects of breast cancer, visit Living with Breast Cancer and Survivorship.

NIH NATIONAL CANCER INSTITUTE

Advances in Breast Cancer Research

NCI-supported researchers are working to advance our understanding of how to prevent, detect, and treat breast cancer. They are also looking at how to address disparities and improve quality of life for survivors of the disease.

This page highlights some of what's new in the latest research for breast cancer, including new clinical advances that may soon translate into improved care, NCI-supported programs that are fueling progress, and research findings from recent studies.

A polyploid giant cancer cell (PGCC) from triple-negative breast cancer.

Credit: National Cancer Institute

On This Page

Research in Breast Cancer Screening

Breast cancer is one of a few cancers for which an effective screening test, mammography, is available. MRI (magnetic resonance imaging) and ultrasound are also used to detect breast cancer, but not as routine screening tools for people with average risk.

Ongoing studies are looking at ways to enhance current breast cancer screening: options:

- The NCI-supported Tomosynthesis Mammographic Imaging Screening Trial (TMIST) is evaluating whether 3-D mammography is better than standard 2-D mammography at detecting breast cancers early enough that they can be treated successfully.

- Several NCI-supported studies are investigating the use of mammography with a contrast dye to improve screening among women with a history of breast cancer and women with dense breasts.
- Some NCI-supported researchers are exploring the use of markers such as breast stiffness to identify cancer in women with dense breasts.
- Researchers are looking for ways to use artificial intelligence (AI) to improve early detection of breast cancer. For example, one NCI-supported study looked at how using artificial intelligence algorithms can improve mammogram interpretation.

Two concerns in breast cancer screening, as in all cancer screening, are:

- the potential for diagnosing tumors that would not have become life-threatening (overdiagnosis)
- the possibility of receiving false-positive test results, and the anxiety that comes with follow-up tests or procedures

To limit the potential for overdiagnosis and overtreatment, researchers are studying screening methods that are appropriate for each woman's level of risk.

The Women Informed to Screen Depending on Measures of Risk (WISDOM) study is comparing risk-based screening—that is, screening at intervals based on an individual's risk—to the standard annual screening mammography.

Researchers are also studying which populations are at highest risk of overdiagnosis. A 2023 study found that older women have a substantial risk of overdiagnosis, highlighting the need for conversations with their health care providers about the potential benefits and harms of continuing screening mammography.

Ductal Carcinoma In Situ

Ductal carcinoma in situ (DCIS) is a precancerous condition in which abnormal cells are found in the lining of a breast duct. Women diagnosed with DCIS have a substantial risk of overdiagnosis and overtreatment because there is currently no way to tell which lesions found on screening mammograms will progress to invasive disease.

Depending on their level of risk, women diagnosed with DCIS almost always have surgery and possibly other treatments. A 2024 trial showed that women with low-risk DCIS who undergo active monitoring were no more likely to be diagnosed with invasive breast cancer after two years than those who undergo surgery with or without radiation therapy. However, longer follow-up is needed to assess the long-term safety of forgoing surgery.

Other trials are ongoing to study whether managing patients diagnosed with DCIS with active surveillance and hormonal therapy can help prevent overtreatment. One NCI-supported study is using artificial intelligence to identify patients with DCIS who are at higher risk of disease progression and may need more aggressive treatment.

Research in Breast Cancer Treatment

The mainstays of breast cancer treatment are surgery, radiation, chemotherapy, hormone therapy, targeted therapy, and immunotherapy. While many drugs have been approved to prevent and treat breast cancer, scientists are continuing to develop new treatments and combinations of treatments to fight the disease.

Hormone Therapy for Breast Cancer

Some breast cancers have receptors for the hormones estrogen and/or progesterone. Therapies that block the growth-promoting effects of estrogen on tumors with hormone receptors can be used to treat such cancers, called hormone receptor (HR), or sometimes estrogen receptor (ER)–positive cancers. Hormone therapy, also referred to as endocrine therapy, can be used to treat both early-stage HR-positive breast cancers and advanced or metastatic disease.

Many hormone therapies exist for treating breast cancer, and these drugs work in several different ways, including suppressing ovarian function, blocking estrogen production, and blocking estrogen's effects.

- A selective estrogen receptor degrader (SERD) is a drug that binds to and breaks down ERs. An oral SERD, called elacestrant dihydrochloride (Orserdu), has recently been approved by the Food and Drug Administration (FDA). It is more effective than other hormone therapies, like fulvestrant (Faslodex), an injectable SERD, for postmenopausal women and men with advanced estrogen receptor (ER)–positive, HER2-negative breast cancer who have mutations in the *ESR1* gene. *ESR1* mutations can make the cancer resistant to other hormone therapies.
- Imlunestrant is also an oral SERD that has been shown to be more effective in slowing the growth of ESR1-mutant tumors in women with advanced, ER-positive, HER2-negative breast cancer than standard hormone therapy. The combination of imlumestrant with

abemaciclib (Verzenio), a drug that blocks cell growth, was better than standard hormone therapy in both those with *ESR1* mutations and those with unmutated *ESR1*.

- In postmenopausal women with hormone receptor-positive early-stage breast cancer, the use of a test that looks at the expression of certain genes can be used to determine whether or not chemotherapy will be beneficial. Ongoing research is looking at whether adding chemotherapy to treatment with ovarian suppression and hormone therapy for premenopausal women with ER-positive/HER2-negative breast cancer reduces the risk of recurrence in women who are at high risk of their cancer returning.
- Researchers are also studying whether the combination of hormone therapy and radiation therapy is more effective than hormone therapy alone in treating women with low-risk early stage breast cancer.
- Tamoxifen, an established hormone therapy for breast cancer, has been shown to reduce the rate of recurrence over 15 years in people with DCIS who were treated without radiation therapy.

Targeted Therapy for Breast Cancer

Many targeted therapies have been approved to treat breast cancer. These targeted therapies, which include monoclonal antibodies and small-molecule drugs, target proteins that control how cancer cells grow, divide, and spread.

Monoclonal antibodies

Monoclonal antibodies are proteins designed to attach to specific targets on cancer cells. One type of monoclonal antibody, called an antibody-drug conjugate, helps carry chemotherapy drugs directly to cancer cells without harming other cells.

Datopotamab deruxtecan (Datroway) is an antibody-drug conjugate under investigation for some metastatic breast cancers, including metastatic triple-negative breast cancers. In a large clinical trial, women treated with datopotamab deruxtecan lived longer without their disease getting worse than women treated with chemotherapy. The drug received FDA approval in 2025 for ER-positive metastatic breast cancer.

- The antibody-drug conjugate trastuzumab deruxtecan (Enhertu) was initially approved for the treatment of HER2-positive breast cancers. The monoclonal antibody trastuzumab (Herceptin) delivers the chemotherapy drug deruxtecan directly to tumor cells that have HER2 on their surface.

- A 2024 study also showed improved outcomes with trastuzumab deruxecan compared to chemotherapy. This led to extend the approval for use in some people with metastatic breast cancer that has low or very low levels of HER2.

Small molecule drugs

Small molecule drugs are a type of targeted therapy that can enter cancer cells and block critical functions.

- In 2024, the FDA approved a triplet therapy that combines the small molecule drug inavolisib (Itovebi) with fulvestrant (Faslodex), a hormone therapy, and palbociclib (Ibrance), another small molecule drug, for certain patients with advanced or metastatic HR-positive breast cancer who have genetic mutations for a protein called PI3K. Inavolisib blocks the resulting overly active forms of PI3K, while palbociclib blocks the activity of the CDK4 and CDK6 proteins, which control cell growth.
- CDK4/6 inhibitors (palbociclib, ribociclib [Kisqali], or abemaciclib [Verzenio]) added to endocrine therapy improve progression-free survival (PFS) in women with metastatic breast cancer. Ribociclib has been shown to improve overall survival (OS).
- For women with early-stage, hormone receptor positive breast cancer, ribociclib given for three years in combination with an aromatase inhibitor has also been shown to reduce risk of recurrence.
- Capivasertib (Trugap), approved by the FDA in 2023, blocks the Akt pathway, which promotes tumor growth. A phase 3 trial showed that capivasertib plus fulvestrant increased how long people with advanced HR-positive breast cancer lived without the disease getting worse.
- For younger women with metastatic ER-positive, HER2-negative breast cancer, results from a large trial showed that the combination of the small molecule targeted drug ribociclib and hormone therapy was much better than chemotherapy at slowing tumor growth.
- In women with high-risk, node positive ER-positive breast cancer, abemaciclib given for two years in combination with endocrine therapy has also been shown to decrease the risk of recurrence.

Immunotherapy for Breast Cancer

Immunotherapy is a type of treatment that helps the body's immune system to fight cancer more effectively. Previous studies have shown that some immunotherapy drugs, known as immune checkpoint inhibitors, improve how long some people with advanced breast cancer can live, particularly those with triple-negative breast cancer. Recent studies provide some evidence that immune checkpoint inhibitors may improve outcomes in some people with HR-positive, HER2-negative breast cancer.

Pembrolizumab (Keytruda) in combination with chemotherapy is approved for treatment of women with metastatic triple negative breast cancer that has high expression of PD-L1. A study in patients with early-stage triple negative breast cancer showed that giving pembrolizumab before surgery, followed by one year of pembrolizumab after surgery, led to better response than chemotherapy alone. This led to FDA approval of pembrolizumab in the early-stage setting.

A current NCI-supported study is looking at patients with early-stage triple-negative breast cancer who had a complete response to treatment before surgery still need more treatment afterward. The standard treatment is to give pembrolizumab 27 weeks after surgery. The goal is to find out if observation alone works just as well as continuing pembrolizumab in preventing the cancer from coming back.

NCI researchers are trying to find ways to use cellular therapies, including CAR T-cell therapy and T-cell transfer therapy, to treat solid tumors, such as breast cancer.

- An ongoing clinical trial is using tumor-infiltrating lymphocytes (TILs) to shrink tumors in women with metastatic breast cancer.
- In another trial, NCI researchers are researching a type of T-cell therapy where a person's T cells are engineered in a laboratory to attack cancer cells and then given back to the patient. The study is recruiting patients with solid cancers, including breast cancer.

For a complete list of drugs for breast cancer, see Drugs Approved for Breast Cancer.

NCI-Supported Breast Cancer Research Programs

Many NCI-supported researchers working at the NIH campus, as well as across the United States and world, are seeking ways to address breast cancer more effectively. Some research is basic, exploring questions as diverse as the biological underpinnings of cancer and the social factors that affect cancer risk. And some are more clinical, seeking to translate this basic information into improving patient outcomes. The programs listed below are a small sampling of NCI's research efforts in breast cancer.

TMIST is a randomized breast screening trial that compares two Food and Drug Administration (FDA)-approved types of digital mammography, standard digital mammography (2-D) with a newer technology called tomosynthesis mammography (3-D).

The Breast Specialized Programs of Research Excellence (Breast SPOREs) are designed to quickly move basic scientific findings into clinical settings. The Breast SPOREs support the development of new therapies and technologies, and studies to better understand tumor resistance, diagnosis, prognosis, screening, prevention, and treatment of breast cancer.

The NCI Cancer Intervention and Surveillance Modeling Network (CISNET) focuses on using modeling to improve our understanding of how prevention, early detection, screening, and treatment affect breast cancer outcomes.

The Confluence Project, from NCI's Division of Cancer Epidemiology and Genetics (DCEG), is developing a research resource that includes data from thousands of breast cancer patients and controls of different races and ethnicities. This resource will be used to identify genes that are associated with breast cancer risk, prognosis, subtypes, response to treatment, and second breast cancers. (DCEG conducts other breast cancer research as well.)

The goal of the Breast Cancer Surveillance Consortium (BCSC), an NCI-supported program launched in 1994, is to enhance the understanding of breast cancer screening practices in the United States and their impact on the breast cancer's stage at diagnosis, survival rates, and mortality.

There are ongoing programs at NCI that support prevention and early detection research in different cancers, including breast cancer. Examples include:

- The Cancer Biomarkers Research Group, which promotes research in cancer biomarkers and manages the Early Detection Research Network (EDRN). EDRN is a network of NCI-supported institutions that are collaborating to discover and validate early detection biomarkers. Within the EDRN, the Breast and Gynecologic Cancers Collaborative Group conducts research on breast and ovarian cancers.
- NCI's Division of Cancer Prevention houses the Breast and Gynecologic Cancer Research Group which conducts and fosters the development of research on the prevention and early detection of breast and gynecologic cancers.

Breast Cancer Survivorship Research

NCI's Office of Cancer Survivorship, part of the Division of Cancer Control and Population Sciences (DCCPS), supports research projects throughout the country that study many issues related to breast cancer survivorship. Examples of studies funded include the impact of cancer and its treatment on physical functioning, emotional well-being, cognitive impairment, sleep disturbances, and cardiovascular health. Other studies focus on financial impacts, the effects on caregivers, models of care for survivors, and issues such as racial disparities and communication.

Breast Cancer Clinical Trials

NCI funds and oversees both early- and late-phase clinical trials to develop new treatments and improve patient care. Trials are available for breast cancer prevention, screening, and treatment.

Breast Cancer Research Results

The following are some of our latest news articles on breast cancer research and study updates:

- Some Women Avoid Breast Cancer Screening After False-Positive Mammogram Results
- Breast Cancer May Spread by Recruiting Nearby Sensory Nerves
- How Breast Cancer Risk Assessment Tools Work
- Can Some People with Breast Cancer Safely Skip Lymph Node Radiation?
- Study Adds to Debate about Mammography in Older Women
- Pausing Long-Term Breast Cancer Therapy to Become Pregnant Appears to Be Safe

View the full list of Breast Cancer Research Results and Study Updates.